

# Outdoor Fix — Milestone 2 & 3 Completion Report

*Developer reference | prepared by Hanzalah*

*Verified live, end to end — not just reviewed in code — on September 1, 2026. One blocking bug and one missing feature were found and fixed in the process.*

## 1. Summary

Milestone 2 (Lease & Ownership Pricing) and Milestone 3 (Risk Assessment) are complete. Every core claim below was tested by actually running the application — submitting a real application as a customer, approving it as an admin, connecting a real Plaid sandbox bank account, and checking the resulting numbers against the formulas in the approved plan — rather than taken on the strength of the code alone.

Two genuine problems surfaced during this pass and were fixed the same session: Plaid bank verification was silently failing for every customer, and the continuous risk red-flag list had never been wired up to anything. Both are detailed in Section 4.

## 2. Milestone 2 — Lease & Ownership Pricing

- **Two ownership paths, one pricing engine.** Full Term Ownership once the Total Rental Purchase Price is paid, and Early Purchase Option pricing that switches formula at the 90-day mark.  
Verified: live EPO figures matched the formula exactly at multiple points across a 36-month term.
- **Automatic payment schedule.** Generated the moment an application is approved — one row per month, correct amount and due date.  
Verified: 36 payment rows created automatically on approval, confirmed in the database.
- **Application wizard, both sides.** The same wizard works for an admin filling it out on a customer's behalf, or for a customer completing it themselves after self-registering.  
Verified: tested a full customer self-service signup → application → submission, separately from the admin-side flow.
- **Live EPO price and payoff chart.** Recalculates immediately as payments post, shown to the customer in the portal.  
Verified: chart and live figure both reflect the current schedule.
- **LDW and Promo code.** Both are tracked per lease agreement and reflected in the payment breakdown when selected.

## 3. Milestone 3 — Risk Assessment

- **Identity, employment, and background checks.** “Verified” means the information is on file, not that a third party has attested to it — this matches the phased scope in the approved plan (see Section 6).
- **Apartment residences auto-decline.** Declined automatically before a reviewer ever sees it.  
Verified: an apartment-residence test application was declined immediately on submission.
- **Affordability check.** An applicant whose reported income falls under the standard 3× monthly-payment ratio is flagged for review.  
Verified: a test applicant reporting \$300/month income against a \$150/month payment was flagged, with the exact reasoning shown on the review screen.
- **Bank verification via Plaid.** Was silently broken for every customer — found and fixed this session, see Section 4.  
Verified: full sandbox connection now completes and shows “verified” with a real timestamp.

- **Post-approval red flags, tracked continuously.** Never wired up to anything before this session — built and verified this session, see Section 4.

## 4. Bugs found and fixed this session

| Issue                   | What was actually happening                                                                                                                                                                                            | Fix                                                                                                                                    |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Plaid bank verification | The plaid_access_token column was a default 255-character field, but Laravel's encrypted cast produces a 300–450 character value. Every real Plaid connection failed with a database error, shown raw to the customer. | Widened the column to TEXT. Re-ran the full connection live — token now saves, account shows “verified.”                               |
| Continuous red flags    | The risk_red_flags table existed in the schema and on the admin screen, but no code anywhere ever wrote a row to it — it was permanently empty for every customer.                                                     | Wired failed payments, overdue payments, and bank-account changes to create real flags, visible and resolvable from the review screen. |

## 5. Verified, not just built

1. A customer self-registers, submits a full application, and it lands in the admin queue with a risk score already attached.
2. An admin reviews, edits terms, and approves — the full payment schedule generates automatically.
3. The EPO figure recalculates correctly at each stage of the term.
4. An apartment-residence application is declined automatically, before it reaches a reviewer.
5. A low-income applicant is automatically flagged, with the reviewer able to see exactly why.
6. A real Plaid sandbox bank account connects successfully end to end, and the connection persists correctly in the database.
7. A failed payment, an overdue payment, and a changed bank account each create a real, visible, resolvable risk flag.

## 6. What's still open

Nothing below is broken — it's the honest scope of what's wired up today, flagged so nobody assumes more coverage than exists yet.

| Area                    | Current state                                                                                                                                                                                             | Status             |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| Bank verification depth | Plaid confirms the account is linked and returns basic account info; deposit history and pay frequency are not pulled yet.                                                                                | Planned, not built |
| Identity / employment   | Self-reported — information is on file, not checked against a government ID service or an employer.                                                                                                       | Planned, not built |
| Red-flag coverage       | failed_ach, missed_payment, and bank_account_change are live. unreachable_customer, suspicious_behavior, undisclosed_move, and the late/missed distinction have no signal in the app yet to trigger them. | Partially built    |

|                      |                                                                                                     |                     |
|----------------------|-----------------------------------------------------------------------------------------------------|---------------------|
| GPS anomaly tracking | Reserved in the schema; live GPS is a Phase 2 addition once the fleet grows, per the approved plan. | Phase 2 (by design) |
| Payments             | Admins mark a payment paid/failed/refunded by hand. No live processor (Stripe) is connected yet.    | Manual for now      |
| Contract generation  | Approving an application generates the payment schedule, not a signed contract.                     | Milestone 5         |

## 7. Next

Milestone 4 — Equipment Tracking: serial-number tracking, service history, and the equipment lifecycle workflow on top of the schema already built.